.NET TECHNOLOGIES
(01CE1523)
Lab Manual

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Batch: B

FACULTY OF ENGINEERING AND TECHNOLOGY
Department of Computer Science and Engineering
01CE1523 - .NET Technologies

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Lab	Program	CO	Date	Marks	Total	Signature

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1.	Develop Student Admission Management Module using OOP Concepts (Class, Object, Constructor, Access Modifiers).	CO1							
2.	Design Employee Payroll System using Inheritance, Interface & Polymorphism	CO1							
3.	Develop Expense Tracking Module with Exception Handling	CO1							
4.	Build Online Event Registration Portal using ASP.NET Controls & Validation	CO2							
5.	Develop Academic Calendar & Leave Management System using Rich Controls, Session & Cookies	CO2							
6.	Create Product Catalog Application using ASP.NET MVC Architecture. (Controllers, Models, Views, Routing)	CO3							
7.	Develop Feedback Management Module using MVC Layout, Partial View & Validation. (HTML Helpers, Validation, Layouts)	CO3							
8.	Develop Inventory Management Module using ADO.NET & CRUD Operations	CO4							

9.	Create Placement Dashboard using Data Controls & Crystal Reports	CO4							
10.	Build Student Management REST API using ASP.NET Core (Middleware, API, Project Structure)	CO5							
11.	Develop Smart Campus Service Portal – Feature Extension (Integrate MVC + Database + API)	CO1 , CO2 , CO3 , CO4 , CO5							

R1	Practical Performance
R2	Timely Submission
R3	Neatness & Documentation
R4	Uploaded on Git

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Experiment 1

AIM: Develop Student Admission Management Module using OOP Concepts (Class, Object, Constructor, Access Modifiers).

Code:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace student_admission_portal .cs
{
 class Student
 {
 private int studentId;
 private string name;
 private int age;
 private int subject1;
 private int subject2;
 private int subject3;
 private int totalMarks;
 private double percentage;
 private string course;
 private string email;
 private string phone;
 private string grade;
 private string collegeEligibility;
 private string courseEligibility;
 private string scholarshipEligibility;
 public Student(int id, string n, int a,
 int s1, int s2, int s3,
 string c, string e, string p)
 {
 studentId = id;
 name = n;
 age = a;

 subject1 = s1;
 subject2 = s2;
 subject3 = s3;

 course = c;
 email = e;
 phone = p;
```





```
 CalculateAdmission();
 }

 private void CalculateAdmission()  {
```

```
totalMarks = subject1 + subject2 + subject3;


percentage = totalMarks / 3.0;

if (percentage >= 90)
{
grade = "A+";
}
else if (percentage >= 80)
{
grade = "A";
}
else if (percentage >= 70)
{
grade = "B";
}
else if (percentage >= 60)
{
grade = "C";
}
else if (percentage >= 50)
{
grade = "D";
}
else
{
grade = "F";
}
if (percentage >= 50 &&
subject1 >= 40 &&
subject2 >= 40 &&
subject3 >= 40)
{
collegeEligibility =
"YES - You are Eligible for College";  }
else
{
collegeEligibility =
"NO - You are Not Eligible for College";  }
```



```
if (collegeEligibility.StartsWith("YES"))
{
if (course.ToLower() == "bca" && percentage >= 50)  {
courseEligibility =
"YES - You are Eligible for BCA";
}
```

```
else if (course.ToLower() == "bba" && percentage >= 45)  {
courseEligibility =
"YES - You are Eligible for BBA";
}
else if (course.ToLower() == "bsc" && percentage >= 55)  {
courseEligibility =
"YES - You are Eligible for BSC";
}
else
{
courseEligibility =
"NO - You are Not Eligible for Selected Course";  }
}
else
{
courseEligibility =
"NO - College Eligibility Required First";  }

if (collegeEligibility.StartsWith("YES"))
{
if (percentage >= 90)
{
scholarshipEligibility =
"YES - You are Eligible for 100% Scholarship";  }
else if (percentage >= 80)
{
scholarshipEligibility =
"YES - You are Eligible for 50% Scholarship";  }
else if (percentage >= 70)
{
scholarshipEligibility =
"YES - You are Eligible for 25% Scholarship";  }
else
{
```



```
scholarshipEligibility =
"NO - You are Not Eligible for Scholarship";
}
}
else
{
scholarshipEligibility =
"NO - You are Not Eligible for Scholarship";
}
}

public void StudentPortal()
```

```
{
Console.WriteLine("\n==========================================");
Console.WriteLine(" STUDENT PORTAL");
Console.WriteLine("==========================================");

Console.WriteLine("Student ID : " + studentId);
Console.WriteLine("Student Name : " + name);
Console.WriteLine("Age : " + age);

// Subject Marks
Console.WriteLine("\n--- SUBJECT MARKS ---");

Console.WriteLine("Subject 1 Marks : " + subject1);
Console.WriteLine("Subject 2 Marks : " + subject2);
Console.WriteLine("Subject 3 Marks : " + subject3);

Console.WriteLine("\nTotal Marks : "
+ totalMarks + " / 300");

Console.WriteLine("Percentage : "
+ percentage.ToString("0.00") + "%");

Console.WriteLine("Grade : " + grade);

// Admission Information
Console.WriteLine("\n--- ADMISSION INFORMATION ---");

Console.WriteLine("Selected Course : " + course);

Console.WriteLine("\nCollege Eligibility:");
Console.WriteLine(collegeEligibility);

Console.WriteLine("\nCourse Eligibility:");
Console.WriteLine(courseEligibility);

Console.WriteLine("\nScholarship Eligibility:");
```



```
Console.WriteLine(scholarshipEligibility);

// Contact Information
Console.WriteLine("\n--- CONTACT INFORMATION ---");

Console.WriteLine("Email : " + email);
Console.WriteLine("Phone : " + phone);

Console.WriteLine("==========================================");  }

public void UpdateCourse(string newCourse)
```

```
{
course = newCourse;

// Recalculate Course Eligibility
CalculateAdmission();

Console.WriteLine("\nCourse Updated Successfully!");
Console.WriteLine("Course Eligibility Updated!");
}

public void UpdateSubject1(int newMarks)
{
subject1 = newMarks;

// Recalculate Everything
CalculateAdmission();

Console.WriteLine(
"\nSubject 1 Marks Updated Successfully!");
}

public void UpdateSubject2(int newMarks)
{
subject2 = newMarks;

// Recalculate Everything
CalculateAdmission();

Console.WriteLine(
"\nSubject 2 Marks Updated Successfully!");
}

public void UpdateSubject3(int newMarks)
{
subject3 = newMarks;
```



```
CalculateAdmission();

Console.WriteLine(
"\nSubject 3 Marks Updated Successfully!");
}

public void UpdateEmail(string newEmail)
{
email = newEmail;

Console.WriteLine(
```

```
"\nEmail Updated Successfully!");
}

public void UpdatePhone(string newPhone)
{
phone = newPhone;

Console.WriteLine(
"\nPhone Number Updated Successfully!");
}

public bool Search(string searchName)
{
return name.Equals(
searchName,
StringComparison.OrdinalIgnoreCase
);
}

public int GetId()
{
return studentId;
}
}

internal class Program
{
static void Main(string[] args)
{
Console.WriteLine("============================================");
Console.WriteLine(
" STUDENT ADMISSION MANAGEMENT SYSTEM");
Console.WriteLine("============================================");
```





```
Console.WriteLine("\n--- STUDENT REGISTRATION ---");

Console.Write("Enter Student ID: ");
int id = Convert.ToInt32(
Console.ReadLine());

Console.Write("Enter Student Name: ");
string name = Console.ReadLine();

Console.Write("Enter Age: ");
```

```
int age = Convert.ToInt32(
Console.ReadLine());

Console.WriteLine(
"\n--- ENTER SUBJECT MARKS ---");

Console.Write("Enter Subject 1 Marks: ");  int
subject1 = Convert.ToInt32(
Console.ReadLine());

Console.Write("Enter Subject 2 Marks: ");  int
subject2 = Convert.ToInt32(
Console.ReadLine());

Console.Write("Enter Subject 3 Marks: ");  int
subject3 = Convert.ToInt32(
Console.ReadLine());


Console.Write(
"\nEnter Course (BCA/BBA/BSC): ");

string course = Console.ReadLine();

// ========================================== //
CONTACT DETAILS
// ==========================================
Console.Write("Enter Email: ");
string email = Console.ReadLine();

Console.Write("Enter Phone: ");
string phone = Console.ReadLine();
```





```
Student student = new Student(
id,
name,
age,
subject1,
subject2,
subject3,
course,
email,
phone
```

```
);

Console.WriteLine(
"\nStudent Registered Successfully!");


Console.WriteLine(
"\nYour Admission Result is Ready!");

student.StudentPortal();

int choice = 0;

while (choice != 9)
{
Console.WriteLine(
"\n==========================================");

Console.WriteLine(
" STUDENT PORTAL MENU");

Console.WriteLine(
"==========================================");

Console.WriteLine(
"1. View Student Portal");

Console.WriteLine(
"2. Update Course");

Console.WriteLine(
"3. Update Subject 1 Marks");

Console.WriteLine(
"4. Update Subject 2 Marks");

Console.WriteLine(
"5. Update Subject 3 Marks");
```



```
Console.WriteLine(
"6. Update Email");

Console.WriteLine(
"7. Update Phone");

Console.WriteLine(
"8. Search Student");
```

```
Console.WriteLine(
"9. Exit");

Console.Write(
"\nEnter your choice: ");

choice = Convert.ToInt32(
Console.ReadLine()


switch (choice)
{
// View Portal
case 1:

student.StudentPortal();

break;

case 2:

Console.Write(
"\nEnter New Course (BCA/BBA/BSC): ");

string newCourse =
Console.ReadLine();

student.UpdateCourse(
newCourse);

// Show Updated Portal
student.StudentPortal();

break;
case 3:

Console.Write(
"\nEnter New Subject 1 Marks: ");
```



```
int newMarks1 =
Convert.ToInt32(
Console.ReadLine());

student.UpdateSubject1(
newMarks1);
```

```
 // Show Updated Portal
student.StudentPortal();

 break;


 case 4:

 Console.Write(
 "\nEnter New Subject 2 Marks: ");

 int newMarks2 =
 Convert.ToInt32(
 Console.ReadLine());

 student.UpdateSubject2(
newMarks2);


 student.StudentPortal();  break;



 case 5:

 Console.Write(
 "\nEnter New Subject 3 Marks: ");

 int newMarks3 =
 Convert.ToInt32(
 Console.ReadLine());

 student.UpdateSubject3(
newMarks3);

 // Show Updated Portal
student.StudentPortal();

 break;
```



```
 case 6:

 Console.Write(
 "\nEnter New Email: ");
```

```
string newEmail =
Console.ReadLine();

student.UpdateEmail(
newEmail)
student.StudentPortal();  break;

case 7:

Console.Write(
"\nEnter New Phone Number: ");

string newPhone =
Console.ReadLine();

student.UpdatePhone(
newPhone);


student.StudentPortal();

break;
case 8:

Console.Write(
"\nEnter Student Name: ");

string searchName =
Console.ReadLine();

if (student.Search(
searchName))
{
Console.WriteLine(
"Student Found!");

student.StudentPortal();  }
Else
```



```
{
Console.WriteLine(
"Student Not Found!");
}

break;
```

```
case 9:

Console.WriteLine(
"\nThank you for using " +
"Student Admission Portal!");

break;


// Invalid Choice
default:

Console.WriteLine(
"\nInvalid Choice! " +
"Please try again.");

break;
}
}

Console.WriteLine(
"\n=============================================");

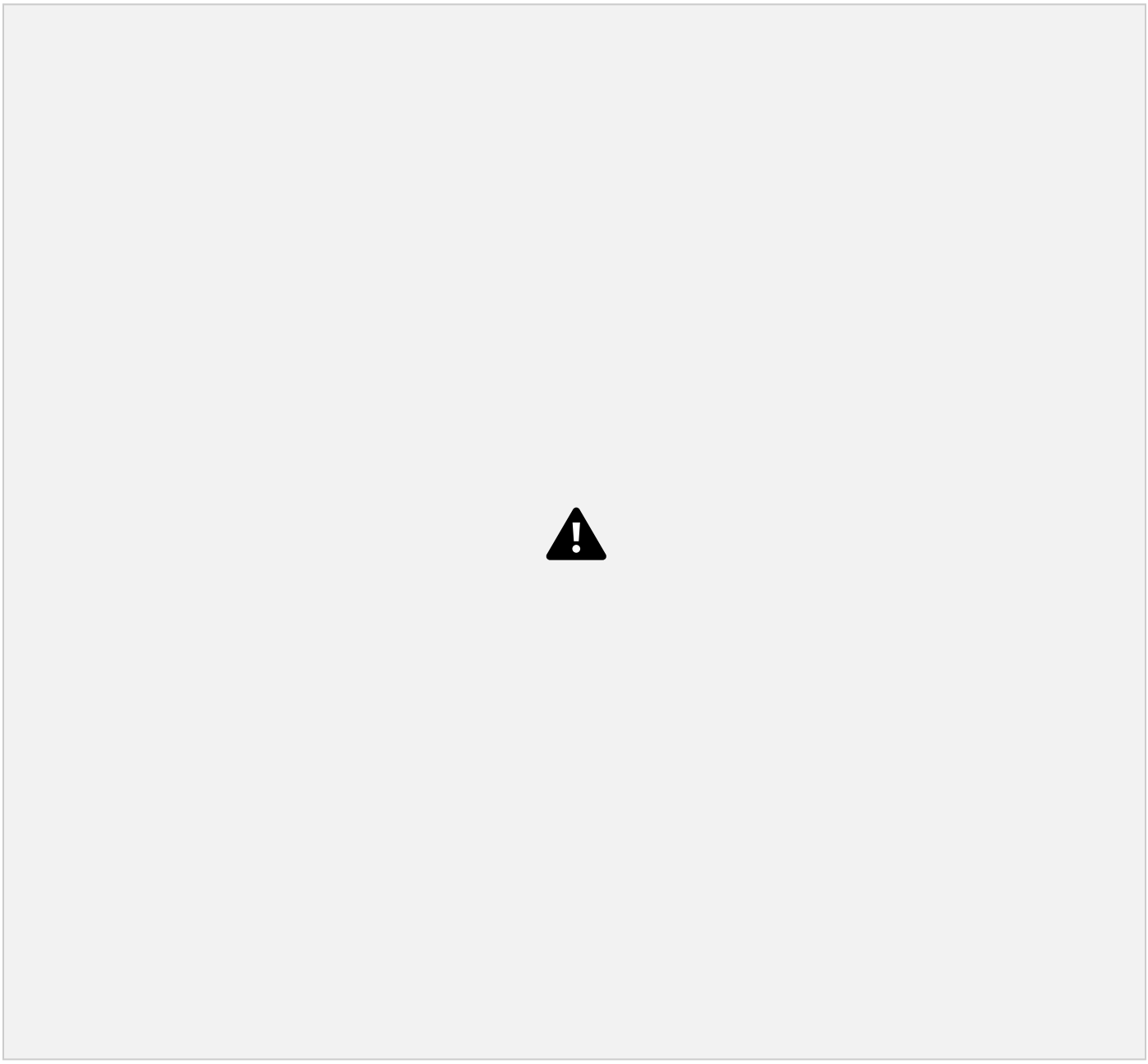
Console.WriteLine(
" ADMISSION SUCCESSFULLY COMPLETED !");

Console.WriteLine(
"=============================================");

Console.ReadLine();
}
}
}
```




Output: Screenshot:



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